Raspberry Pi 5 Homelab: Audio Pipeline, Alarm Architecture & Troubleshooting Guide

Complete Reference • BlueZ Bluetooth Pairing • MPD & PulseAudio Routing • Home Assistant One UI Alarms & Automation Failsafes

1. Linux Command-Line Audio & Bluetooth Infrastructure

This section covers pairing your Bluetooth speaker via SSH/PowerShell and configuring the lightweight Linux audio daemon (MPD) to route sound through PulseAudio with full software volume control.

A. Bluetooth Pairing via SSH / Command Line (BlueZ)

To pair and connect your **HAVIT MX701** (MAC: `17:18:07:A9:99:2D`) or any external Bluetooth audio receiver:

```
# 1. Enter interactive Bluetooth control prompt
sudo bluetoothctl

# 2. Inside bluetoothctl, turn on power, agent, and scanning:
power on
agent on
default-agent
scan on

# 3. Locate your speaker MAC address in the list, then pair, trust, and connect:
pair 17:18:07:A9:99:2D
trust 17:18:07:A9:99:2D
connect 17:18:07:A9:99:2D

# 4. Exit interactive prompt
exit
```

B. Music Player Daemon (/etc/mpd.conf) — Software Mixer & PulseAudio Routing

To ensure MPD runs under your user account and allows Home Assistant 100% volume control without throwing `No mixer` errors, edit `sudo nano /etc/mpd.conf`:

```
# 1. Run under your user account (allows access to paired Bluetooth sockets)
user            "redwannabil"

# 2. Configure PulseAudio output with software mixer enabled
audio_output {
        type            "pulse"
        name            "HAVIT MX701 Bluetooth"
        server          "/run/user/1000/pulse/native"
        mixer_type      "software"
}
```

After saving, fix permissions for your user account and restart the MPD daemon:

```
sudo chown -R redwannabil:audio /var/lib/mpd /var/log/mpd
sudo systemctl restart mpd
```

C. MPD Command-Line Control & Testing (mpc)

Verify that your siren audio files play directly to your speaker from the terminal:

```
# Check active MPD playback status and volume
mpc status

# List available alarm files inside /var/lib/mpd/music/alarms/
mpc ls alarms

# Clear queue, load alarm siren, force 100% volume, and start playback
mpc clear && mpc add "alarms/nuclear_siren.mp3" && mpc volume 100 && mpc play
```

2. Spotify Connect & YouTube Music Integration Architecture

How to stream external music services through your Raspberry Pi and HAVIT MX701 speaker with near-zero CPU overhead (< 35 MB RAM).

A. Raspotify (Spotify Connect Linux Endpoint)

Raspotify turns your Raspberry Pi into an official Spotify Connect speaker controlled from any Spotify mobile/PC app:

```
# 1. Install Raspotify official repository
curl -sL https://dtcooper.github.io/raspotify/install.sh | sh

# 2. Edit configuration: sudo nano /etc/raspotify/conf
LIBRESPOT_NAME="HAVIT MX701 Soundbox"
LIBRESPOT_BACKEND="pulseaudio"

# 3. Restart service
sudo systemctl restart raspotify
```

B. YouTube Music Integration (ytube_music_player) & Cookie JSON File

To prevent `async_get_cipher failed` and bot-detection rate limits, store your YouTube Music authentication cookie inside a JSON file on your Pi: `sudo nano /home/redwannabil/homeassistant/ytmusic_headers.json`:

```
{
  "User-Agent": "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/127.0.0.0 Safari/537.36",
  "Accept": "*/*",
  "Accept-Language": "en-US,en;q=0.5",
  "Content-Type": "application/json",
  "X-Goog-AuthUser": "0",
  "x-origin": "https://music.youtube.com",
  "Cookie": "YSC=PASTE_YOUR_FULL_COOKIE_HERE; VISITOR_INFO1_LIVE=..."
}
```

INTEGRATION SETUP PATH

In Home Assistant (**Settings** → **Devices & Services** → **YouTube Music Player**), enter the file path `/config/ytmusic_headers.json` in the UI configuration (never paste raw cookie strings into the file path box) and select `media_player.music_player_daemon` as the default output.

3. Kobold Alarm & Production-Ready Home Assistant Automations

This complete `automations.yaml` block includes crash protection (`continue_on_error: true`), infinite siren looping (`repeat: "one"`), queue replacement (`enqueue: replace`), interactive phone notifications, and a Bluetooth keep-alive heartbeat.

```yaml
# ==============================================================================
# 1. MAIN ALARM TRIGGER (100% Volume + Infinite Loop + Phone Push Notification)
# ==============================================================================
- id: 'kobold_alarm_wakeup_master'
  alias: "Kobold Alarm: Max Volume Ringtone Wake Up"
  description: "Loops siren endlessly at 100% volume and sends actionable notification"
  mode: restart
  trigger:
    - platform: state
      entity_id: input_boolean.kobold_alarm_trigger
      to: "on"
  action:
    - action: media_player.turn_on
      target:
        entity_id: media_player.music_player_daemon
      continue_on_error: true
    - action: media_player.volume_set
      target:
        entity_id: media_player.music_player_daemon
      data:
        volume_level: 1.0
      continue_on_error: true
    - action: media_player.repeat_set
      target:
        entity_id: media_player.music_player_daemon
      data:
        repeat: "one"
      continue_on_error: true
    - action: media_player.play_media
      target:
        entity_id: media_player.music_player_daemon
      data:
        media_content_type: music
        media_content_id: "alarms/nuclear_siren.mp3"
        enqueue: replace
      continue_on_error: true
    - action: notify.notify
      data:
        title: "ALARM CLOCK RINGING!"
        message: "Your wake-up alarm is sounding on the HAVIT MX701 speaker!"
        data:
          actions:
            - action: "ALARM_DISMISS"
              title: "Dismiss Alarm"
            - action: "ALARM_SNOOZE"
              title: "Snooze (10 Mins)"

# ==============================================================================
# 2. ALARM STOP (Silences Speaker on Dismiss)
# ==============================================================================
- id: 'kobold_alarm_stop_master'
  alias: "Kobold Alarm: Stop Audio on Dismiss"
  trigger:
    - platform: state
      entity_id: input_boolean.kobold_alarm_trigger
      to: "off"
  action:
    - action: media_player.media_stop
      target:
        entity_id: media_player.music_player_daemon
      continue_on_error: true

# ==============================================================================
# 3. PHONE NOTIFICATION CONTROLS (Handles Tapping Dismiss or Snooze on Phone)
# ==============================================================================
```

```yaml
- id: 'kobold_alarm_phone_control_master'
  alias: "Kobold Alarm: Mobile Phone Notification Control"
  mode: parallel
  trigger:
    - platform: event
      event_type: mobile_app_notification_action
      event_data:
        action: "ALARM_DISMISS"
      id: "dismiss"
    - platform: event
      event_type: mobile_app_notification_action
      event_data:
        action: "ALARM_SNOOZE"
      id: "snooze"
  action:
    - choose:
        - conditions:
            - condition: trigger
              id: "dismiss"
          sequence:
            - action: input_boolean.turn_off
              target:
                entity_id: input_boolean.kobold_alarm_trigger
        - conditions:
            - condition: trigger
              id: "snooze"
          sequence:
            - action: media_player.media_stop
              target:
                entity_id: media_player.music_player_daemon
              continue_on_error: true
            - action: input_boolean.turn_off
              target:
                entity_id: input_boolean.kobold_alarm_trigger
            - delay: "00:10:00"
            - action: input_boolean.turn_on
              target:
                entity_id: input_boolean.kobold_alarm_trigger

# ==============================================================================
# 4. BLUETOOTH KEEP-ALIVE HEARTBEAT (Prevents Speaker Deep Sleep)
# ==============================================================================
- id: 'system_bluetooth_heartbeat_master'
  alias: "System: HAVIT MX701 Keep-Alive Heartbeat"
  mode: restart
  trigger:
    - platform: time_pattern
      minutes: "/10"
  action:
    - action: media_player.volume_set
      target:
        entity_id: media_player.music_player_daemon
      data:
        volume_level: 1.0
      continue_on_error: true
```

4. Samsung One UI Masonry Dashboard Architecture

To avoid bulky cards and "custom element doesn't exist" Lovelace errors, keep your main dashboard set to **Masonry** or **Sections** view mode (never Panel mode) and use standalone vertical stacks.

A. Clean One UI Audio & Alarm Control Card (Lovelace YAML)

```yaml
type: vertical-stack
cards:
  - type: custom:mushroom-title-card
    title: "Home Soundbox"
    subtitle: "HAVIT MX701 Audio System"

  # 1. COMPACT ALARM TILE (Tap opens details subview)
  - type: tile
    entity: input_boolean.kobold_alarm_trigger
    name: "Wake-Up Alarm"
    icon: mdi:alarm
    color: blue
    tap_action:
      action: navigate
      navigation_path: /lovelace/alarm-details

  # 2. UNIFIED PLAYER CONTROL
  - type: media-control
    entity: media_player.music_player_daemon

  # 3. DIRECT RADIO & LOFI BEAT BUTTONS
  - type: horizontal-stack
    cards:
      - type: button
        name: "24/7 Lofi Beats"
        icon: mdi:headphones
        tap_action:
          action: media_player.play_media
          target:
            entity_id: media_player.music_player_daemon
          data:
            media_content_type: music
            media_content_id: "http://ice1.somafm.com/groovesalad-128-mp3"
            enqueue: replace
      - type: button
        name: "Stop Audio"
        icon: mdi:stop-circle-outline
        tap_action:
          action: media_player.media_stop
          target:
            entity_id: media_player.music_player_daemon
```

B. Kobold Alarm Details Subview Card (/lovelace/alarm-details)

In your dedicated alarm subview page, set `show_clock: false` to hide the giant digital clock font and display a clean One UI schedule list:

```yaml
type: custom:kobold-alarm-clock-card
title: "Samsung One UI Alarms"
alarm_entities:
  - input_boolean.kobold_alarm_trigger
show_clock: false
time_format: 12
show_next_alarm: true
show_days: true
show_controls: true
```

5. Complete Master Troubleshooting & Maintenance Reference Table

Use this reference table to immediately diagnose and resolve any audio, alarm, or streaming issue from scratch.

ERROR / SYMPTOM	ROOT CAUSE & IMMEDIATE SCRATCH FIX
`MPD error: No mixer volume: n/a`	MPD is outputting via ALSA directly or lacks a software mixer. Edit `/etc/mpd.conf` to include `mixer_type "software"` inside the PulseAudio block and run `sudo systemctl restart mpd`.
`can't find header file at YSC=...`	You pasted the raw cookie string into the integration's file path box. Store the cookie inside `/home/redwannabil/homeassistant/ytmusic_headers.json` and enter `/config/ytmusic_headers.json` in the UI.
`Please select a player before start playing (/)`	The internal HACS controller lost its target speaker binding. Add an explicit `media_player.select_source` step in your automation targeting `media_player.ytubemusic_player` with source `"Music Player Daemon"`.
`CommandError: [50@0] {load} No such playlist`	MPD tried to read an external YouTube playlist URL as a local Linux `.m3u` file. For account playlists, call `media_player.play_media` on `media_player.ytubemusic_player` (NOT directly on MPD).
`This live stream recording is not available`	24/7 YouTube live streams require a Deno/Node JS runtime installed for `yt-dlp`. Use direct MPD radio streams (e.g., `http://ice1.somafm.com/groovesalad-128-mp3`) or standard uploaded YouTube songs instead.
`Unable to find Kobold card in lovelace config`	The Kobold Alarm Clock card was nested inside a `vertical-stack`. Place the Kobold card as a standalone top-level card on your dashboard view.
`Speaker asleep / silent first few seconds`	PulseAudio suspended the idle Bluetooth sink. Add `media_player.turn_on` before playback and run the 10-minute heartbeat automation to prevent deep sleep.

MASTER AUDIO PIPELINE DIAGNOSTIC COMMAND

To verify Linux permissions, MPD socket routing, and Bluetooth speaker playback in one single terminal line, run: `mpc clear && mpc add alarms/nuclear_siren.mp3 && mpc volume 100 && mpc play`